

Invers Matriks 2 x 2

Rumus ringkas dan contoh perhitungan

Jika $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, maka

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \text{ dengan } \det(A) = ad - bc \neq 0$$

Contoh

A

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\det(A) = 1(4) - 2(3) = -2$$

$$A^{-1} = \frac{1}{-2} \times \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$$

B

$$B = \begin{bmatrix} -3 & 5 \\ 2 & -2 \end{bmatrix}$$

$$\det(B) = (-3)(-2) - 5(2) = -4$$

$$B^{-1} = \frac{1}{-4} \times \begin{bmatrix} -2 & -5 \\ -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 5/4 \\ 1/2 & 3/4 \end{bmatrix}$$

C

$$C = \begin{bmatrix} -3 & -2 \\ 4 & 3 \end{bmatrix}$$

$$\det(C) = (-3)(3) - (-2)(4) = -1$$

$$C^{-1} = \frac{1}{-1} \times \begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & -2 \\ 4 & 3 \end{bmatrix}$$

Catatan penting

- Invers hanya ada jika determinan $\neq 0$.
- Tukar posisi a dan d.
- Ubah tanda b dan c.
- Kalikan dengan $1/\det(A)$.

Determinan Matriks 3 x 3

Ekspansi kofaktor baris pertama

Misalkan $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Pola tanda kofaktor baris pertama: $\quad \quad \quad + \quad - \quad +$

Contoh

$$D = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$$

$$\begin{aligned} \det(D) &= 1[(4)(6) - (5)(0)] \\ &\quad - 2[(0)(6) - (5)(1)] \\ &\quad + 3[(0)(0) - (4)(1)] \\ &= 1(24) - 2(-5) + 3(-4) \\ &= 24 + 10 - 12 \\ &= \mathbf{22} \end{aligned}$$

Cara cepat mengingat

1. Pilih satu baris/kolom.
2. Tutup baris dan kolom elemen.
3. Hitung determinan minor 2 x 2.
4. Ikuti pola tanda +, -, +.

Cek hasil: determinan D = 22, sehingga D memiliki invers.